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Techno-Economic Analysis of Solar-Wind Hybrid Systems for Green Hydrogen Production in South Africa, KwaZulu-Natal Province

Sandile Mtolo[1] | Sudesh Rathilal[1] | Nomcebo Happiness Mthombeni[1] | Katleho Moloi[2] | Emmanuel Kweinor Tetteh[1]

[1]Green Engineering Research Group, Durban University of Technology, Musgrave, South Africa | [2]Department of Power Engineering, Durban University of Technology, Durban, South Africa

Correspondence: Sandile Mtolo (21960172@dut4life.ac.za) | Emmanuel Kweinor Tetteh (emmanuelk@dut.ac.za; ektetteh34@gmail.com)

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ABSTRACT

Rising global energy demand and the climate crisis have intensified the pursuit of renewable energy, with green hydrogen emerging as a key alternative to fossil fuels. South Africa, heavily reliant on coal-based energy, faces pressure to decarbonize and transition to a green economy. In this study, the techno-economic feasibility of producing green hydrogen via seawater electrolysis with the Durban climatic conditions for a hybrid solar-wind battery system. Hybrid optimization of multiple energy resources (HOMER) was used to simulate and compare three configured systems (PV, wind and hybrid PV-wind battery) on net present cost (NPC), levelized cost of hydrogen (LCOH), and levelized cost of energy (LCOE). The hybrid system proved most effective, yielding LCOH of \$8.94/kg and 45.8 tons/year of hydrogen, outperforming the standalone systems. With 97.5% excess electricity generated, the system supports scalability for grid integration, positioning green hydrogen as a viable, cost-effective path toward a sustainable energy future for South Africa.

1 | Introduction

Global energy demand is rising abruptly because of high population growth and industrialization and is anticipated to grow by 28% through 2040 [1]. Fossil fuels, currently the main source of energy, are a significant driver of global warming and ecological devastation. Consequently, there is an increasing transition to renewable energies and cleaner energy carriers such as hydrogen, which is known to have high specific energy and efficiency, especially as a storage medium [1, 2]. Hydrogen, which can be derived from several resources such as coal, natural gas, and biomass, is also being generated in an environmentally friendly way through water electrolysis powered by renewable energy systems [1, 3]. This transition aims to limit environmental impacts and provide a sustainable future in energy. The use of hydrogen in the renewable energy transition is most promising because it can be utilized as a universal energy carrier. Solar photovoltaic (PV) and wind turbines (WT) are gaining popularity as renewable energy technologies to provide the world's energy needs.

Rising global energy demand and the climate crisis have intensified efforts to find renewable energy sources (RESs), with green hydrogen becoming a major alternative to fossil fuels. Green hydrogen, produced via electrolysis powered by renewable sources, plays a crucial role in decarbonizing energy systems and achieving net-zero emissions [4, 5]. South Africa, which relies on coal-based energy, is under pressure to switch to a green economy, making this study both timely and relevant.

However, the intermittent characteristics of renewable energy resources, fueled by unpredictable weather conditions, necessitates cost-effective and reliable energy storage methods [1]. In off-grid systems, where the generation may exceed 60% of the overall production, the utilization of hydrogen generation systems can be particularly beneficial. Excess renewable energy can power electrolyzers to produce green hydrogen, which can be stored and, at a later time, converted back to electricity by utilizing fuel cells, providing backup power in low-generation cycles [6]. This method is especially beneficial to remote areas, where grid expansion is economically and logistically challenging [6].

South Africa, despite having reasonably affordable electricity costs, has huge energy access issues, especially in rural areas. By 2020, around 8 million people had no access to electricity [7]. People had no access to electricity [4]. The country's abundant solar resources render solar PV as a promising option, though at high costs that prevent its extensive use [8]. To address these issues, the South African government has launched initiatives such as HYSA (Hydrogen South Africa), aimed at deploying hydrogen and fuel cell technologies to enhance energy access, particularly in off-grid and rural settings [3, 9]. Green hydrogen, produced from water electrolysis using renewable energy, is considered a key decarbonization solution for the energy system and sustainable energy access [9].

The transition to eco-friendly transport options is crucial to tackle environmental issues such as carbon emissions and climate change. Among these options, hydrogen refueling stations (HRS) integrated with RES have shown promise especially in rural areas with limited energy access [10]. Safak and Çiçek's (2025) research focuses on optimizing grid-connected HRS in rural areas for multiple objectives showing how RES integration can benefit both the economy and environment, which aligns with the goals of this study. South Africa, which relies on coal-based energy, faces similar challenges in decarbonizing its transportation sector, making green hydrogen a possible solution.

The commitment to curb global warming under the Paris Agreement [11] has also increased the focus on clean energy carriers like hydrogen. Green hydrogen, generated through electrolysis from renewable energies, will be instrumental in the attainment of net-zero emissions, with production projected to grow from 0.5 million tons in 2020 to 80 million tons in 2030 [12]. Countries rich in renewable resources, such as South Africa, are well-positioned to lead the green hydrogen market both domestically and in terms of exports. However, challenges remain, particularly in the requirement for purified water in current electrolyzer technologies. In water-scarce regions like South Africa, seawater can be used as a potential feedstock for the production of hydrogen, but desalination steps would be required to meet the low conductivity requirements of electrolyzers [13]. Table 1 outlines major green hydrogen production studies, with gaps that this paper bridges. Most studies were on inland renewable hydrogen systems based on freshwater electrolysis, with little consideration of coastal hybrid solar-wind configurations or seawater electrolysis. Although some studies were using HOMER for techno-economic optimization, none of them specifically concentrated on South Africa's KwaZulu-Natal (KZN) coast, where there is high renewable potential but little consideration in hydrogen research. Additionally, earlier research used to concentrate on either cost (NPC, LCOH) or energy efficiency (LCOE) in isolation, as opposed to integrating all three with the inclusion of seawater electrolysis as a green alternative, critical for water-scarce coastal areas.

The simulation of green hydrogen production through seawater electrolysis in KZN, South Africa, using the HOMER software represents a critical step toward realizing the potential of renewable energy-driven hydrogen economies. South Africa, and particularly KZN, has prospects for green hydrogen production from hybrid solar-wind systems, and there are research gaps to be filled. Research focuses on optimal PV, WT, and electrolyzer sizing complemented with battery storage for resilience. The region's coastal resources may enable scalable low-carbon hydrogen production. Global green hydrogen prices remain high (~$5/kg) [24], while declining renewable costs may improve feasibility. South Africa's renewable resource tracks global trends but requires localized techno-economic assessment [25].

Recent studies by Çiçek have highlighted the importance of hybrid renewable systems for green hydrogen production, near coastlines where using seawater for electrolysis can help solve freshwater shortages [10]. The integration of solar and wind resources not only enhances system reliability but also supports global efforts to cut carbon emissions, as demonstrated in resilience-oriented energy management strategies for green buildings [10]. Furthermore, Safak and Çiçek [14] demonstrated trade-offs between cost minimization and load factor maximization by optimizing a grid-interactive HRS with RES, achieving hydrogen costs of $3.03/kg and financial gains up to $587.83.

Erding et al. [17] have explored resiliency-sensitive decision-making mechanisms for residential communities with hybrid energy systems, including hydrogen storage and fuel cell electric vehicles (FCEVs). Their study focuses on making the power grid more resilient and multi-energy systems. However, these studies primarily focus on urban or inland applications, leaving a gap for coastal regions with seawater electrolysis, as addressed in our work. Our research, study uniquely addresses the techno-economic feasibility of seawater electrolysis for green hydrogen production in a coastal African context, using a hybrid of solar and wind power. This distinction highlights how our work contributes to finding local solutions for water-scarce regions.

A study on analysis of advanced green H_2 production methods was conducted by Naqvi et al. [5]. Their findings revealed the following costs for green hydrogen production methods: PEM electrolysis ($2.94–3.32/kg, $600/kW capex), biomass gasification ($3.47–4.11/kg, 94.9–99.1 kg H_2/ton biomass), and thermochemical water-splitting ($3.92/kg, 168.3 kg/h yield). Their study also emphasized cost reductions via hybrid systems (e.g., PV-wind-electrolysis).

Recent advancements in desalination-integrated hydrogen production by Romo et al. [15] demonstrate the feasibility of coastal hybrid systems. These systems reach a 47% recovery ratio with minimal energy loss and have less than 10% error in simulated measurements. Their study reported 45.8 tons of hydrogen per year at $8.94/kg (LCOH), proving that seawater

TABLE 1 | Summary of techno-economic studies from different regions.

Reference	Location/year	Main content (research approach and observation)
Safak and Çiçek [14]	Süloğlu, Türkiye (2025)	Optimized a grid-interactive hydrogen refueling station with RES, achieving hydrogen costs of \$3.03/kg and financial gains up to \$587.83. Demonstrated trade-offs between cost minimization and load factor maximization.
Rebeka Beres et al. [12]	South Africa (2024)	Analyzed feasibility of 500 kt annual hydrogen production by 2030. Found costs (\$2–3/kg) competitive but emission factors (13–24 kg CO_2/kg H_2) exceeded EU benchmarks, necessitating massive RES capacity expansion.
Çiçek [10]	Nottingham, UK (2024)	Proposed a resilience-oriented energy management strategy for hydrogen-based green buildings. Achieved 29% cost reduction using RES and FC, with hydrogen support from HEVs during outages. Highlighted feasibility of zero-carbon operation.
Ahmad et al. [4]	Global (2024)	Reviewed various hydrogen production technologies, noting coal gasification (50–150 g H_2/kg feedstock, \$0.92–2.83/kg, 57%–60% efficiency) and steam methane reforming (\$0.77–2.75/kg, 70%–85% efficiency) dominate current production. Highlighted biomass gasification (\$1.21–3.5/kg, 25%–30% efficiency) and electrolysis (high purity but costly, \$2.53–2.80/kg for bio-photolysis) as emerging alternatives.
Naqvi et al. [5]	Global (2024)	Analyzed advanced green H_2 production methods: PEM electrolysis (\$2.94–3.32/kg, \$600/kW capex), biomass gasification (\$3.47–4.11/kg, 94.9–99.1 kg H_2/ton biomass), and thermochemical water-splitting (\$3.92/kg, 168.3 kg/h yield). Emphasized cost reductions via hybrid systems (e.g., PV-wind-electrolysis).
Romo et al. [15]	University of Maryland, USA (2024)	Proposed a novel modeling method for multieffect distillation (MED) and reverse osmosis (RO) desalination plants with energy recovery. Achieved simulation errors of 2.5%–9.3% compared to actual plant data. Introduced a theoretical efficiency metric to identify underperforming plants, with some operating 20% below optimal recovery. Highlighted the role of energy recovery devices in reducing energy intensity.
Ravi Samikannu et al. [16]	Botswana (2022)	Designed PV/wind/battery systems for rural electrification, noting economic challenges due to high initial costs but potential for cost reductions as RES prices decline globally.
Fatma Gülgen Erding et al. [17]	Istanbul, Turkey (2022)	Developed a resiliency-sensitive energy management system for a residential community with fuel cell electric vehicles (FCEVs), PV, and hydrogen storage. Achieved cost savings (e.g., −1263.81 TL profit in Case-1) and minimized curtailment (0 kWh in Case-2). Demonstrated the effectiveness of common energy storage (CESS) and FCEVs in reducing grid dependency during outages.
T.R. Ayodele et al. [18]	South Africa (2021)	Conducted techno-economic analysis of wind-powered hydrogen refueling stations, identifying coastal cities as optimal locations. Reported hydrogen costs of \$6.34/kg–\$8.97/kg and significant CO_2 reductions.
Kakoulaki et al. [19]	EU and UK (2021)	Highlighted RES potential to meet electricity and hydrogen demand, with 81% of regions achieving surplus RES generation. Aligned with EU decarbonization goals for hydrogen integration.
Jahangiri et al. [20]	Qatar (2020)	Assessed home-scale solar-wind systems for electricity (14 kWh) and hydrogen (85 kg/day) production. Found Grid-connected scenarios offered the lowest hydrogen cost (\$2.09/kg) and PV-Wind-Grid configurations minimized CO_2 emissions (1434 kg annually).

(Continues)

TABLE 1 | (Continued)

Reference	Location/year	Main content (research approach and observation)
Gokcelek and Kale [21]	Turkey (2018)	Analyzed hybrid renewable systems for hydrogen refueling stations, reporting production costs ranging from \$7.53/kg to \$11.08/kg. Demonstrated regional viability of hydrogen production using RES.
Al-Buraiki and Al-Sharafi [22]	Dhahran, Saudi Arabia (2017)	Evaluated a hybrid PV-wind-battery system using hourly simulations, achieving an LCOE of \$0.593/kWh and hydrogen production cost of \$36.32/kg, with annual CO_2 reduction of 9.66 tons. Highlighted the feasibility of hybrid systems for hydrogen production.
Siyal et al. [23]	Sweden (2015)	Evaluated wind-powered hydrogen production, achieving costs between \$5.18/kg and \$9.62/kg. Emphasized the role of turbine selection in cost variability.

Abbreviations: CO_2, carbon dioxide; EU, European Union; FC, fuel cell; HEV, hybrid electric vehicle; LCOE, levelized cost of energy; PEM, proton exchange membrane; PV, photovoltaic; RES, renewable energy sources.

electrolysis can scale up to address freshwater shortages, a key point to consider for South Africa's renewable energy transition.

Ahmad et al. [4] reviewed various hydrogen production technologies, noting coal gasification (50–150 g H_2/kg feedstock, \$0.92–2.83/kg, 57%–60% efficiency) and steam methane reforming (\$0.77–2.75/kg, 70%–85% efficiency) dominate current production. Their study also highlighted biomass gasification (\$1.21–3.5/kg, 25%–30% efficiency) and electrolysis (high purity but costly, \$2.53–2.80/kg for bio-photolysis) as emerging alternatives.

This study addresses critical research gaps by focusing on green hydrogen production in the coastal region of KZN, where location-specific analyses and hybrid solar-wind systems remain underexplored, leveraging HOMER Pro 3.14.2 (developed by the U.S. National Renewable Energy Laboratory, NREL, and distributed by UL Solutions, Boulder, CO) to conduct a techno-economic optimization. The optimization focused on minimizing the net present cost (NPC), levelized cost of hydrogen (LCOH), and Levelized Cost of Energy (LCOE), while also advancing the understanding of seawater electrolysis, a less studied but promising alternative to freshwater electrolysis, particularly for water-scarce coastal regions.

The rest of the paper is organized as follows: Section 2 describes the data and methodology approach. Section 3 discusses the results and lastly, Section 4 summarizes the conclusions and recommendations for future work.

2 | Methodology

2.1 | Site Description

This study assumes that the proposed site is in a coastal region of KZN, with coordinates approximately −30.134735° latitude and 30.83725° longitude, and this green hydrogen production system is designed to meet a hydrogen demand of 100 kg/day. Figure 1 shows the site location of the proposed HRES (Hybrid Renewable Energy Systems). Umgababa is located about 36 km south of Durban, lying between Illovo Beach and Umkomaas. It's along the same stretch of coast as Amanzimtoti, a popular seaside destination. The site of the proposed green hydrogen plant is said to be in Southern Umgababa It has an area of 6.23 km² and a population of 10,814, with a population density of 1,735.80 inhabitants per km² [26].

The coastal area KZN Province was selected due to its proximity to the sea, which facilitates seawater electrolysis for green hydrogen production. The area's renewable energy (RE) potential was evaluated by Marc-Alain Mutombo and Numbi [27], who assessed the available RE resources in the KZN province. Their findings indicate that the province has a total RE potential of approximately 45 GW, distributed as follows: 53.63% from global horizontal irradiance (GHI), 23.28% from direct normal irradiance (DNI), 13.52% from wind energy, 9.51% from geothermal energy, and 0.06% from biomass. Ocean energy and hydropower were excluded from this assessment [27]. Even though analysis conducted by Mutombo and Numbi [27] conclusively identified Nonoti as the site with the highest solar energy potential, the site has the highest GHI measurement at 1679.7 kWh/m², the strongest DNI reading at 1647.8 kWh/m², and the most impressive P_{VOUT} value of 1511.1 kWh/kWp, While the selected KZN coastal (Umgababa) area is the third largest with GHI of 1613.3 kWh/m^2, DNI of 1538.9 kWh/m^2 and PVOUT of 1463.2 kWh/kWp at an optimum tilt angle of 31° [27]. The highest wind speeds in KZN occur in the eastern and western regions, exceeding 4 m/s, making them suitable for wind farms and energy generation. Areas with wind speeds of 3–4 m/s are moderate. Wind energy potential in KZN is estimated at 6167 GW, covering 13.30% of the province's area [27]. However, their study revealed that, in terms of hydro potential, Umgababa Dam site is the second-best site for the inclusion of hydropower development. The reservoir of the site has a reservoir capacity (m³) of 1,280,000 m^3 following Hazelmere Dam (17,860,000 m^3) which is distant from the sea [27]. The location is assumed to be strategically positioned to leverage these RESs, with a consistent hydrogen load profile across all locations.

2.2 | Components of the Renewable Hydrogen Facilities System and Input Parameters

2.2.1 | Description of the Hybrid Renewable Power Generation Systems and Components

In this study, an optimization and techno-economic analysis of a Solar-Wind Hybrid Renewable Energy System for Green

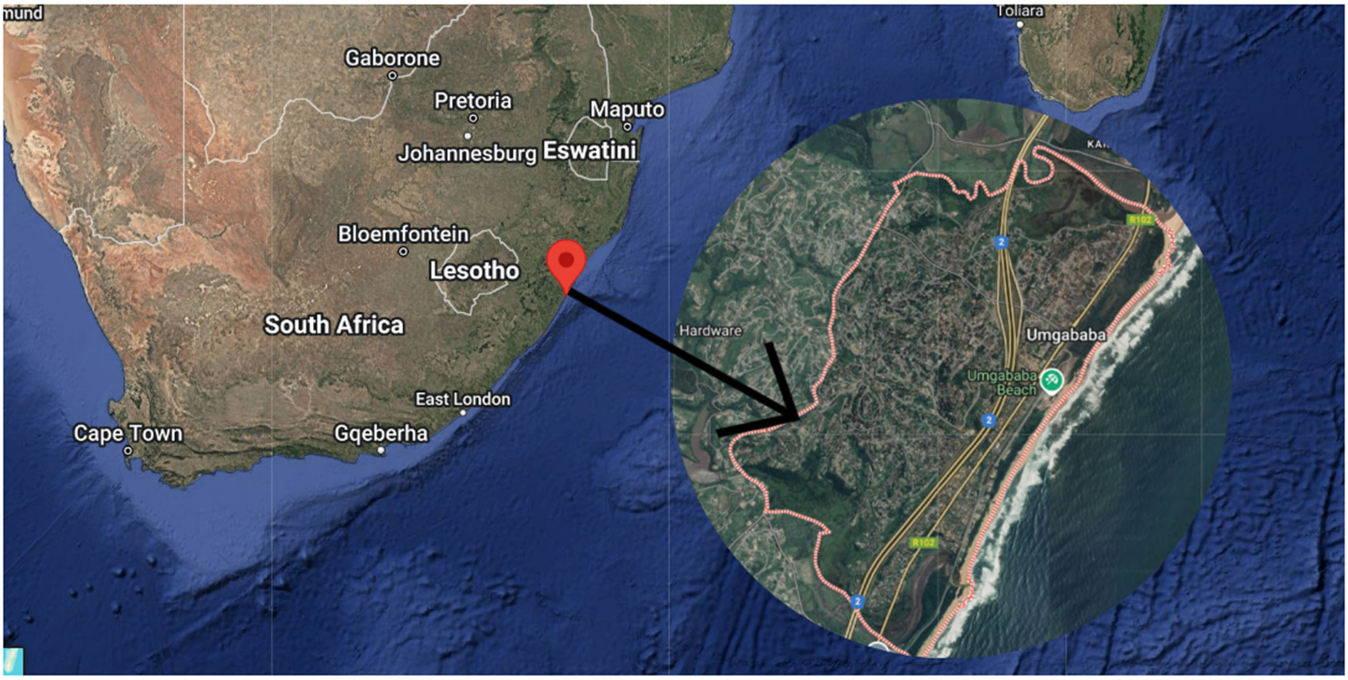

FIGURE 1 | Site Location, Umgababa, Durban, KwaZulu Natal, South Africa (Google Earth).

Hydrogen Production via Seawater Electrolysis in was carried out using the HOMER software (version 3.14.2). This software was initially created by the National Renewable Energy Laboratory (NREL), a division of the U.S. Department of Energy, and is distributed by UL Solutions, a company headquartered in Boulder, Colorado, USA. Renewable energy systems like solar and wind are subject to inconsistency due to weather conditions. Solar generation ceases at night and during heavy cloud cover, while WTs stop producing power if wind speeds drop below their cut-in threshold (typically 3–4 m/s). On the other hand, very high winds (above 25 m/s) may trigger a cut-off for safety reasons. Dependence on a single renewable source often necessitates oversizing system components, driving up costs and reducing efficiency [22]. A hybrid system combining solar panels and WTs can reduce component sizes and energy production costs. A hybrid solar/wind system proposed in this paper is meant to meet two demands: the daily electricity needs of residential homes and the hydrogen demand. The system prioritizes electricity supply, and excess energy, after charging batteries, is used to produce hydrogen via an electrolyzer. The hydrogen is stored in tanks and used to meet hydrogen demand [22].

The hybrid solar-wind system has an influence on complementary generation profiles (solar peaks during daylight and wind peaks at night) to make sure there's a stable energy supply for electrolysis, which helps to optimize hydrogen output [4]. This approach aligns with global trends where hybrid systems are prioritized for their resilience and cost-effectiveness [5]. Furthermore, this approach is also supported by Safak and Çiçek [14], who demonstrated that RES integration improves economic feasibility by reducing costs and increasing financial gains while enhancing system resilience. The hybrid system can balance out the intermittency in energy production. This aligns with the multi-goal optimization framework proposed in their study, which aims to maximize both cost efficiency and load factor performance.

As depicted in Figure 2, the green hydrogen renewable energy system consists of the following components: (1) Renewable power generation system (PV-battery (Case 1), wind-battery (Case 2), and hybrid PV-wind-battery (Case 3), (2) PEM electrolyzer (hydrogen production system), (3) hydrogen storage tank to safely store the produced hydrogen.

HOMER Software is widely recognized for its robust economic and technical models, which incorporate detailed component and economic analyses. HOMER Pro identifies energy systems with the lowest NPC that meet specific energy demands under defined constraints. The input includes technical and economic parameters related to the technologies under consideration, renewable energy resources (wind, solar, temperature, and so on), and economic parameters relevant to the project (discount rate, inflation rate, project life, and so on). The software operates by balancing energy production and consumption at hourly intervals over a year, estimating installation and operational costs for feasible configurations throughout the project's lifetime. The optimizer program of HOMER Pro finds the best configuration of the system, including the converters, storage systems, PV arrays, and WTs, that minimizes the NPC. For this study, unmet hydrogen load was set to 0% to ensure 100% coverage of the yearly demand by the renewable hydrogen facility [18]. The optimizer was used to determine the optimal sizing of each component within the green hydrogen production system.

2.2.2 | Hydrogen Profile

The load profile of the hydrogen production plant in the proposed location is shown in Figure 3. HOMER software is a commercial tool specifically designed to optimize the configurations of microgrid power systems and therefore, it is suitable to optimize the size of each component in the hybrid renewable

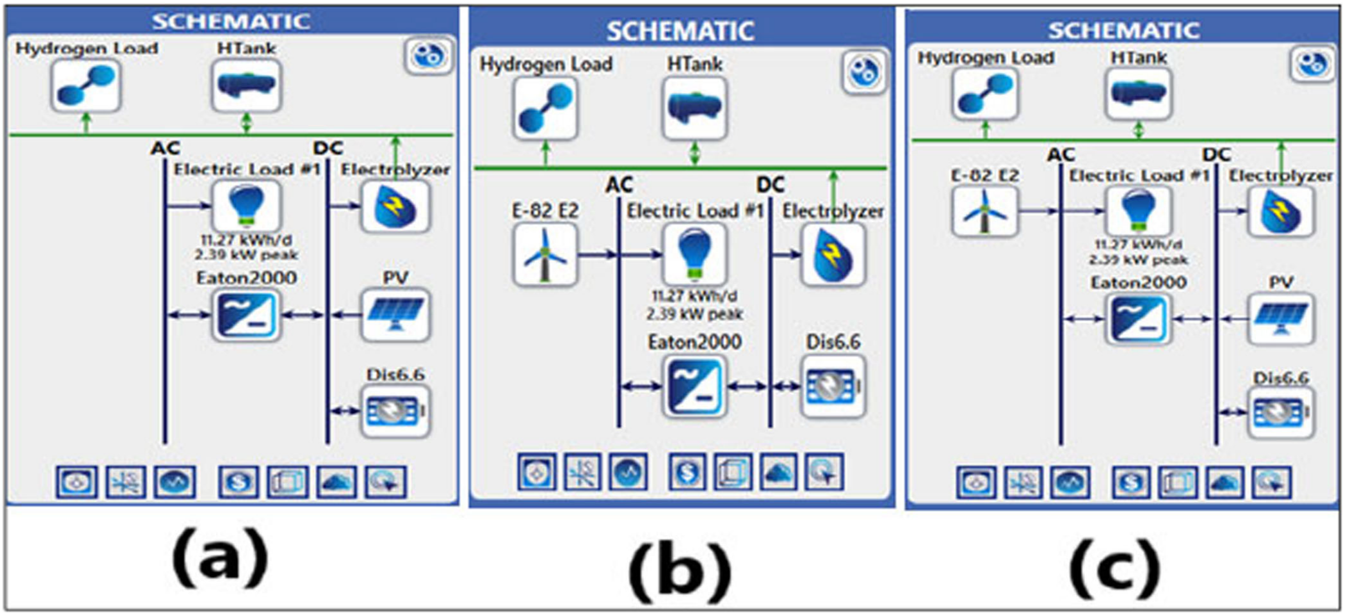

FIGURE 2 | HOMER optimization simulation design models, (a) PV system only, (b) Wind system only, and (c) Hybrid PV-Wind System.

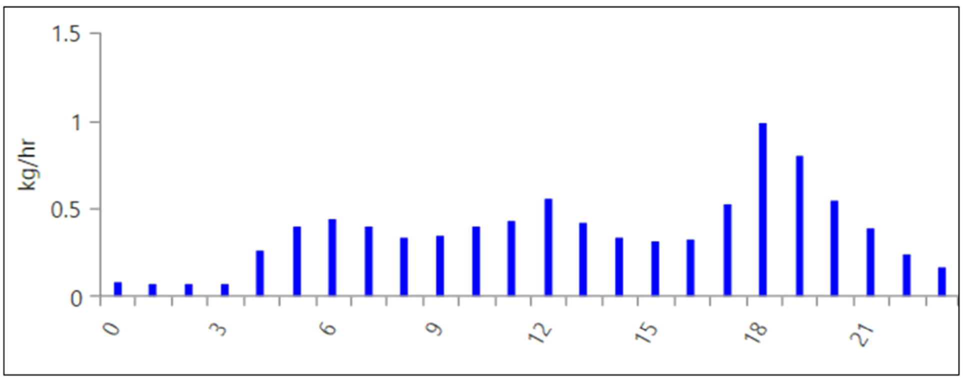

FIGURE 3 | Hourly hydrogen load profile in KwaZulu-Natal for year 2021 obtained from the HOMER software.

energy systems by techno-economic analysis. The unmet hydrogen load (which indicates the hydrogen load that cannot be met by the hydrogen production system throughout the year 2021) was set at 0% for the simulations. Figure 4 shows the HOMER simulation design flowchart detailing the steps involved in the techno-economic analysis.

The HOMER software optimization flowchart (Figure 4) illustrates a systematic approach to designing renewable energy systems. It begins with the determination of meteorological information and geographical location and then detailing system load and energy generating components. This involves defining technical specifications, conducting simulations, and evaluating design results through optimum component sizing, costing analysis, and technical analysis, with the target of determining the most favorable setup of the renewable energy system [28].

The meteorological data for solar resources was sourced from the National Aeronautics and Space Administration (NASA) within the hybrid optimization model for electric renewables (HOMER) software tool were used to estimate the electrical energy generated by the renewable power generation systems. The solar radiation data and clearness index values were used as the inputs of the simulations using HOMER software, as shown in Figure 5a. The solar radiation varies from 3.14 to 5.31 kWh/m^2/day at the Umgababa site, with an annual average daily value of 4.41 kWh/m^2/day. The average clearness index is 0.52 based on the monthly values. The solar radiation is highest in January and December, whereas the solar radiation is lowest in June. Figure 5b shows that the wind speed is within a range of 5.51–6.84 m/s, measured at an anemometer height of 10 m. The wind speed is highest in October and the average annual wind speed at the Umgababa site is 6.14 m/s. The monthly average daily temperature profile is also shown in Figure 5c. The temperature varies from 16.79°C to 23.19°C, the temperatures are the highest in February and the lowest in July.

2.3 | System Elements and Their Design Principles

This section outlines the components of the system and the foundational principles guiding its design. The analysis begins with an examination of the mathematical model equations and the technical specifications of each system component. Subsequently, the

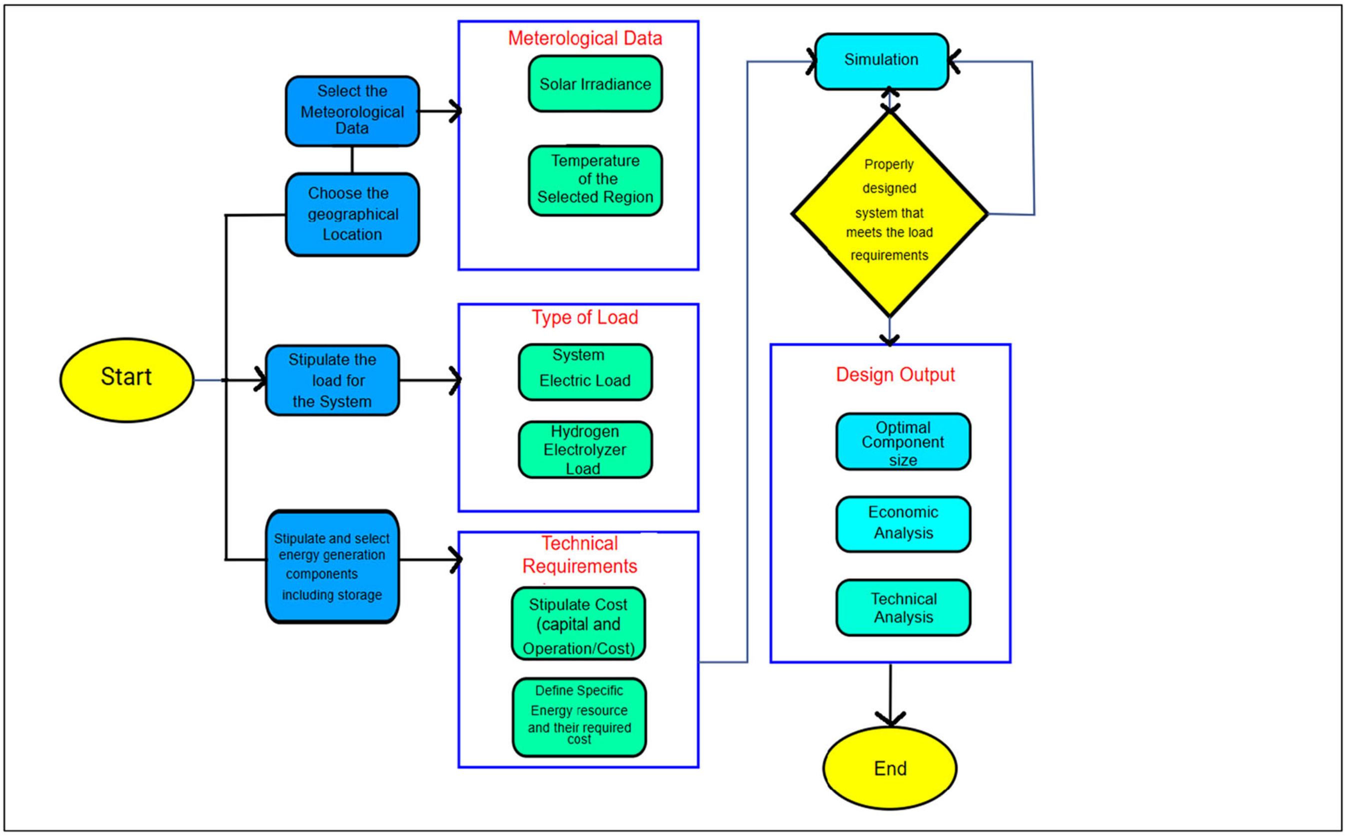

FIGURE 4 | HOMER simulation design flowchart [28].

costs associated with the essential machinery and provisions will serve as the basis for the economic evaluation.

2.3.1 | WT Model

The power curve of a WT represents the relationship between wind speed at hub height and the corresponding power output. HOMER calculates turbine output power using linear interpolation based on the entered power curve data. Power generation is zero at wind speeds outside the curve's range, as the turbine shuts down when speeds are either too low to produce energy or too high, risking damage [29]. In this study, the Enercon E-82 E2 turbine, with a 2 MW capacity and 85 m hub height, was utilized as depicted in Figure 6. Wind speed data were sourced from NASA's surface meteorology and solar energy database, built in HOMER and also from [30]. The monthly average wind speed profile at a 10 m anemometer height is shown in Figure 5b.

The WT's electricity output was calculated through a systematic process. First, the wind speed at the turbine's hub height was estimated using the logarithmic law. Next, the generated electricity was determined by integrating the turbine's characteristic power curve with wind speed and standard air density data measured at the hub height. Finally, the output power was adjusted to reflect the actual air density [29]. The wind speed at the hub height was calculated using the following power law:

$$U_{\text{hub}} = U_{\text{anem}} \left(\frac{Z_{\text{hub}}}{Z_{\text{anem}}} \right)^{\beta}. \tag{1}$$

The wind speed at the turbine hub height (U_{hub}) is calculated using the wind speed at the anemometer height (U_{anem}) and the exponential factor (β), which depends on surface roughness and atmospheric stability. The factor β typically ranges between 0.05 and 0.5. Previous research indicates that β varies between 0.104 and 0.196 for various sites in South Africa [18]. For this study, an average value of 0.1 was adopted.

The WT output was calculated by applying a correction factor to account for the actual air density, as given by Equation (2) [29].

$$P_{\text{WT}} = \left(\frac{\rho}{\rho_0} \right) \cdot P_{\text{WT,STP}}. \tag{2}$$

Here, ρ represents the air density in kilograms per cubic meter (kg/m^3), and ρ_0 denotes the air density under standard conditions (kg/m^3). PWT_STP is the WT's output power in kilowatts (kW) under standard conditions, derived from the manufacturer's characteristic power curve. This study selected the ENERCON E-82 E2 WT, with a rated power of 2 MW.

2.3.2 | PV Array Modeling

A PV array was used as part of the off-grid hybrid renewable power generation system to supply electricity to the PEM electrolyzer. The output power of the PV array was determined using Equation (3) as follows:

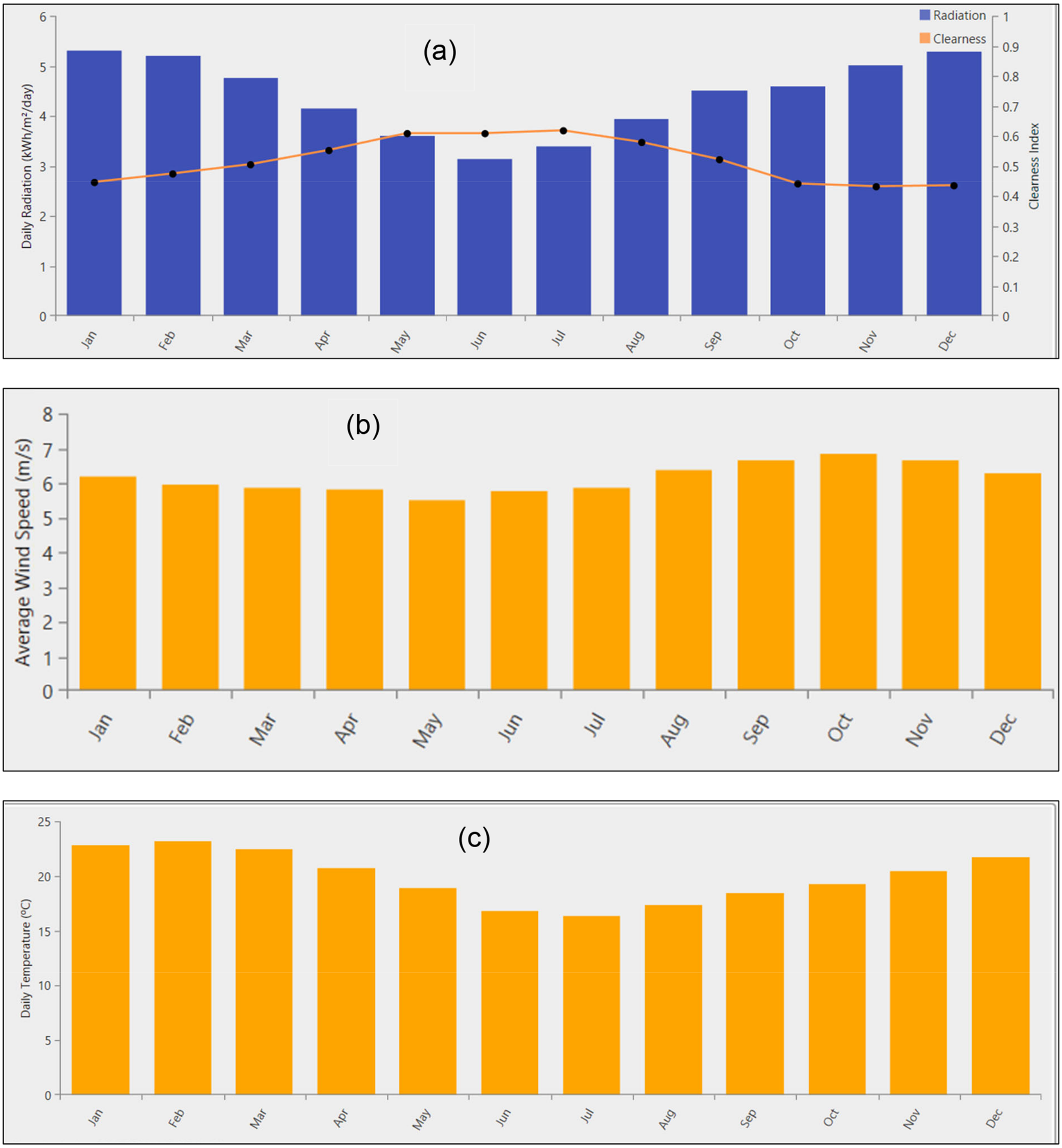

FIGURE 5 | (a) Average monthly solar GHI, (b) Average monthly wind speed, (c) Average monthly temperature in Umgababa, Durban, KwaZulu-Natal, South Africa in 2021.

$$P_{\mathrm{PV\_OUTPUT}} = P_{\mathrm{PV\_RATED}}\, F_{\mathrm{PV}} \left(\frac{G_T}{G_{T,\mathrm{STC}}} \right). \tag{3}$$

The rated capacity of the PV array, $P_{\mathrm{PV\_rated}}$, is expressed in kilowatts (kW), where F_{PV} represents the derating factor (%). The derating factor in this study is set to 80% [21]. G_T denotes the solar radiation incident on the PV array (W/m^2), and GT_{STC} is the standard test condition value of 1 kW/m^2. Temperature effects on PV panel performance were excluded from the analysis. The PV array capacity was determined using HOMER's auto-sizing generator. All PV panels were modeled at a fixed tilt angle of 30° without a tracking system.

2.3.3 | Electrolyzer (PEMEL)

Electrolyzers are devices that use electrical energy to split water into hydrogen and oxygen through electrolysis. They are

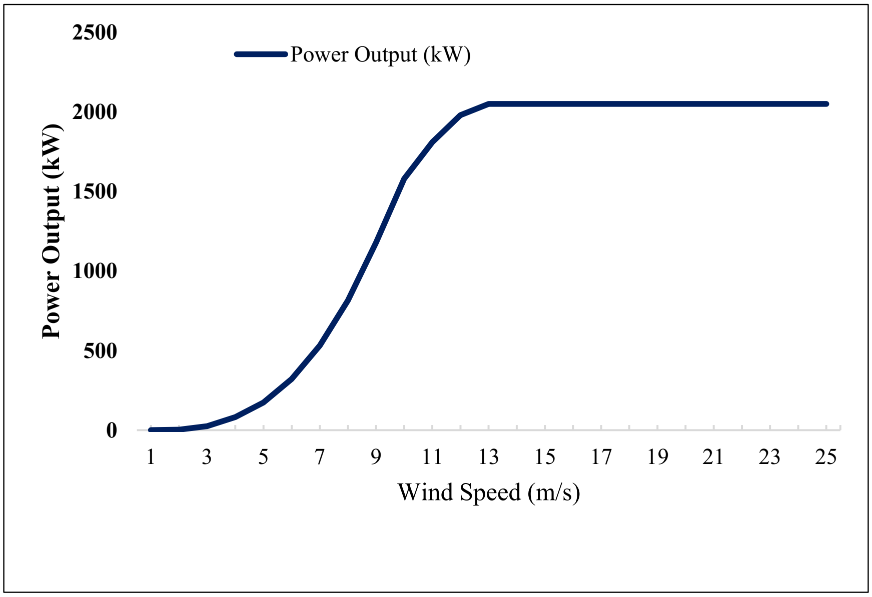

FIGURE 6 | Wind power curve of the Enercon E-82 E2 wind turbine.

primarily categorized into two types: alkaline electrolyzers and proton exchange membrane (PEM) electrolyzers. Alkaline electrolyzers typically operate at temperatures between 60°C and 80°C, with conventional systems functioning at pressures of around 5 bar, while advanced systems can achieve pressures of 10–30 bar [18, 22]. In contrast, PEM electrolyzers represent a more recent technological advancement, addressing limitations of alkaline systems such as low current density, partial load performance, and low-pressure operation. PEM electrolyzers are characterized by their rapid dynamic response, broad operational range, high efficiency (exceeding 85%), and exceptional gas purity levels (up to 99.999%) [18].

PEM electrolyzers utilize a proton-conducting membrane, typically made of Nafion, as a solid electrolyte. The design is simpler compared to alkaline systems, and they theoretically achieve efficiencies of up to 94% based on the higher heating value (HHV), with practical efficiencies exceeding 85% [21, 31]. During electrolysis, water is split into hydrogen and oxygen. Hydrogen is produced at the cathode (Equation 4), while oxygen is generated at the anode (Equation 5) [31]:

$$2H^{+} + 2e^{-} \rightarrow \quad H_2, \tag{4}$$

$$H_2O \quad \rightarrow \quad \frac{1}{2}O_2 + 2H^{+} + 2e^{-}. \tag{5}$$

PEM electrolyzers are particularly advantageous for small-scale energy generation in remote areas due to their high efficiency and compatibility with renewable energy systems [29, 31]. The efficiency of PEM electrolyzers is often assumed to be around 80% based on HHV in practical applications [16, 31].

The mass flow rate of hydrogen produced by an electrolyzer can be calculated using the following Equation (6) [31].

$$\dot{m}_{H_2}(t) = \frac{3600 \cdot P_{el}(t) \cdot \eta_{\mathrm{el}}}{\mathrm{HHV}_{\mathrm{H_2}}}, \tag{6}$$

where:

- $\dot{m}_{H_2}(t)$ is the mass flow rate of hydrogen (kg/h),
- $P_{\mathrm{el}}(t)$ is the electrical power input to the electrolyzer (kW),
- η_{el} is the electrolyzer efficiency,
- HHV_{H_2} is the HHV of hydrogen (39.44 kWh/kg or 142 MJ/kg).

For example, a 1 kW PEM electrolyzer with 74% efficiency can produce approximately 0.0187 kg/h of hydrogen [71]. According to the Department of Energy (DOE), an efficiency of 87% is considered optimal for PEM electrolyzers [21].

PEM electrolyzers are increasingly favored over alkaline systems due to their operational simplicity, superior efficiency, and enhanced hydrogen production capabilities [18]. They are particularly suitable for integration with renewable energy systems, such as solar and wind, making them ideal for sustainable power generation frameworks [18]. In South Africa, PEM electrolyzers benefit from the abundant availability of platinum, a key material used in their electrodes [3, 18]. In addition to their operational advantages, PEM electrolyzers utilize platinum as an electrode material, which is abundantly available in South Africa. This makes PEM electrolyzers a practical choice for green hydrogen production in the region of Umgababa location.

The rate of hydrogen production (α_{H2}) in a PEM electrolyzer can be calculated using the following equation [18]:

$$\alpha_{H_2} = \frac{h_f \cdot N_s \cdot I_{\mathrm{el}}}{\eta \cdot F}, \tag{7}$$

where:

- h_f is the Faraday efficiency,

- Ns is the number of cells in series,
- I_{el} is the electrolyzer current,
- η is the number of electrons per mole of hydrogen (typically 2),
- F is Faraday's constant (96,485,309 C/mol).

In HOMER Pro, the electrolyzer is modeled as a component rather than as a load. The configuration page for an electrolyzer in HOMER Pro includes capital, replacement, and O&M costs by capacity, enabling optimization of electrolyzer sizing as part of the system design. The electrolyzer operates flexibly, receiving excess electricity for hydrogen production while also demanding electricity when hydrogen storage is insufficient, aligning with component-based modeling logic.

2.3.4 | Hydrogen Tank

Hydrogen tanks are critical for storing hydrogen produced by electrolyzers, ensuring a consistent supply to meet demand. The accumulated hydrogen in the tank at any hour (t) is determined by the difference between the produced hydrogen and the hydrogen load, as described by Equation (8) [22].

$$\mathrm{CHT}\quad(t) = \mathrm{CHT}\quad(t-1) + \dot{m}_{H_2}(t) - \dot{L}_{H_2}(t), \tag{8}$$

where CHT(t) is the hydrogen tank capacity at hour (t), $\dot{m}_{H_2}(t)$ is the mass flow rate of hydrogen produced, and $\dot{L}_{H_2}(t)$ is the hydrogen load. The tank capacity is constrained by minimum and maximum limits [22]:

$$\mathrm{CHT}\quad(t)_{\min} \leq \mathrm{CHT}\quad(t)\quad \leq \mathrm{CHT}\quad(t)_{\max}, \tag{9}$$

where $\mathrm{CHT}_{\min}$ and $\mathrm{CHT}_{\max}$ represent the minimum and maximum allowable capacities, respectively. For large-scale stationary storage, tank size is optimized with a typical useful life of 25 years [32]. The mass flow rate of hydrogen into the tank, $Q_{H2}(t)$, is calculated as [31]:

$$Q_{H_2}(t) = \dot{m}_{H_2}(t) - \dot{f}_{H_2}(t) - \dot{L}_{H_2}(t), \tag{10}$$

where $\dot{f}_{H_2}(t)$ is the input mass flow rate to the fuel cell.

The amount of hydrogen added to the tank within an hour, $T_{\text{level}}\ (t)$, is given by [31]:

$$T_{\text{level}}(t) = Q_{H_2}(t). \tag{11}$$

The updated tank level, $T_{\text{level,new}(t)}$, is calculated as [31]:

$$T_{\text{level,new}}(t) = \sum_{t=1}^{t=t-1} T_{\text{level}}(t) + T_{\text{level}}(t), \tag{12}$$

where the first term represents residual hydrogen and the second term accounts for hydrogen stored in the current hour. High-pressure gas cylinders are the most widely used storage technology, with recent advancements enabling lightweight composite cylinders to withstand pressures up to 800 bar, achieving a volumetric density of 36 kg/m³ [18]. The pressure inside the tank can be determined using [18]:

$$Q_{H_2} = \frac{R \cdot T}{V_{\text{ht}}} \cdot n_{\text{ht,}} \tag{13}$$

where V_{ht} is the tank volume, n_{ht} is the number of moles of hydrogen, R is the gas constant, and T is the temperature. Hydrogen tanks store excess hydrogen during periods of low demand or surplus production, assuming no leakage [18].

Hydrogen storage faces challenges due to its low volumetric density (0.089886 kg/m³ at 20°C and 1 atm), necessitating compression or cooling. Advanced composite cylinders, such as the 80 MPa tanks with carbon fiber shells and thermoplastic liners, are now preferred for their high-pressure tolerance and efficiency. The volume of compressed hydrogen can be calculated using [18]:

$$Q_{H_2} = \frac{M_{H_2}}{h_{\text{cm}} \cdot s_c}, \tag{14}$$

where M_{H2} is the mass of hydrogen, s_c is the compressed hydrogen density (36 kg/m³ at 800 bar and 25°C), and h_{cm} is the compressor efficiency (0.95 for low-speed reciprocating compressors, chosen for their high efficiency and suitability for high compression ratios).

2.3.5 | Storage Batteries

The intermittent nature of RESs necessitates the use of storage batteries to ensure a continuous electricity supply for hydrogen production. When the load demand is lower than the electricity generated by the hybrid power system, excess energy from WTs or PV arrays can charge the batteries until they reach full capacity [18]. In this study, the Discover AES 6.6 kWh, 48 Vdc battery was selected for energy storage due to its high round-trip efficiency of approximately 95% and a lifetime of 38,000 kWh over 10 years [29]. The capacity of each battery was determined using the auto-sizing generator in the HOMER software. The system stores the generated energy in the battery and delivers it to the load during power outages. The overall efficiency η is calculated as [33]:

$$\eta_{\text{overall}} = \eta_{\text{inv}}\eta_{\text{rect}}\eta_{\text{rt}}, \tag{15}$$

where

- η_{overall} (Overall Efficiency): The total efficiency of the entire system, accounting for energy losses across all stages (conversion, storage, and delivery).
- η_{inv} (Inverter Efficiency): The efficiency of the inverter, which converts DC power (e.g., from batteries or solar panels) to AC power for the load. Typical values range from 90% to 98%.

- η_{rect} (Rectifier Efficiency): The efficiency of the rectifier, which converts AC power (e.g., from the grid or generator) to DC power for battery storage. Typical values range from 85% to 95%.
- η_{rt} (Round-Trip Efficiency): The efficiency of the battery storage system, accounting for energy losses during charging and discharging. For lithium-ion batteries, this is typically 85%–95%.

Formula (15) is critical for designing reliable power systems (e.g., solar and storage, UPS) where minimizing energy losses ensures optimal performance during outages [33].

2.3.6 | Converter

A converter is used to manage energy flow between the DC and AC buses. The Eaton Power Xpert 2000 kW, a bidirectional, three-phase converter with an inverter and rectifier efficiency of 98%, was employed in the simulation. The optimal sizes of both the battery and converter were determined using the optimizer tool in HOMER [29].

Alternating Current (AC) to Direct Current (DC) Conversion

The conversion of AC to DC involves rectification and relies on the root mean square (RMS) voltage calculation. For a sinusoidal AC waveform, the RMS voltage represents the equivalent DC voltage that would deliver the same power [34].

Key Equation:

$$\mathrm{RMS} = \frac{V_P}{\sqrt{2}}, \tag{16}$$

where V_p is the peak AC voltage.

This relationship allows for bidirectional calculations:

a. **AC to DC:** For an AC peak voltage of 141 V:

$$100\mathrm{VDC} = \frac{141}{\sqrt{2}}. \tag{17}$$

b. **DC to AC:** To power a 100 V DC device, the required peak AC voltage is:

$$V_P = 100 \quad \sqrt{2} = 141. \tag{18}$$

This study considers a general system converter with an efficiency of 95%. The converter efficiency is calculated using Equation (19).

$$\eta_{\mathrm{conv}} = \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}}. \tag{19}$$

2.4 | Techno-Economic Analysis

The economic feasibility of an HRES powered by a wind-PV hybrid system with battery backup was evaluated using HOMER software. This tool assesses the system's technical performance and its ability to meet energy demands efficiently [21]. The economic feasibility analysis relies on key indicators such as the NPC, LCOE, and the LCOH production, ranking optimization results from the lowest to the highest cost. HOMER computes these metrics by evaluating the present value of capital costs, replacement costs, maintenance, and operational expenses over the system's lifetime.

2.4.1 | NPC

The present value of revenues is then subtracted from these costs, as outlined in Equation (15) [35].

$$\mathrm{NPC} = \frac{C_{\mathrm{ann,tot}}}{\mathrm{CRF}(i, N)}, \tag{20}$$

where

$C_{\mathrm{ann,tot}}$ is the total annualized cost ($/year).

CRF represents the capital recovery factor.

N is the life cycle period in years.

i is the yearly real discount rate and can be determined using Equation (3)

$$i = \frac{i^o - f}{1 - f}, \tag{21}$$

where i^o is the nominal discount rate and f is the expected inflation rate. In this study, the discount rate and inflation rate were taken as 5.9% and 1.5% [36], respectively. These values were taken from the data published by the South African Reserve Bank in 2025 [36]. The project lifetime was assumed to be 25 years. Table 2 shows a list of the parameter values used in the economic analysis of the system [18].

The CRF (capital recovery factor) was calculated using Equation (6)

$$\mathrm{CRF} = \frac{i(1+i)^n}{(1+i)^n - 1}. \tag{22}$$

2.4.2 | LCOE

The LCOE is defined as the average cost per kWh of useful energy produced by the power-producing system [29] (i.e., the PV- wind hybrid system). The LCOE can be determined using:

$$\mathrm{LCOE} = \frac{C_{\mathrm{ann,tot}}}{E_{\mathrm{served}}}, \tag{23}$$

where

- $C_{\mathrm{ann,tot}}$ is the total annualized cost ($/yr),
- E_{served} is load served (kWh/yr).

TABLE 2 | The techno-economic data input for the power systems considered [18].

Component	Capital cost ($/kW)	Replacement ($/kW)	Operations and maintenance cost ($/kW/year)	Life cycle (Year)	Capacity (kW or kg)
Generic flat Plate PV	2500	2500	10	25	1
Wind turbine (Enercon E-82)	1000	1000	50	20	2000
Electrolyzer (PEM)	1500	1000	20	15	300
Hydrogen tank	665	400	10	25	500 kg
Battery (Discovery AES)	1200	1000	10	10	6.6 kWh
Eaton power Xpert-converter	1000	1000	10	15	100

An estimate of the techno-economical feasibility of hybrid systems involves a comparison of the annual cost to demand met. The annual cost is the sum of the annualization of costs of each component of the hybrid energy system. A critical consideration in conducting this calculation is the energy consumption of the electrolyzer. The HOMER software, commonly used for such analyses, does not account for the electric load consumed by the electrolyzer when calculating the LCOE [29]. This is because HOMER categorizes the electrolyzer as a component rather than a load. To address this limitation and ensure a more accurate LCOE calculation, the total annual electrical energy consumed by the electrolyzer was included in the total annual electrical load in this study [18, 29].

The electrical energy consumed by the electrolyzer per kilogram of hydrogen produced was determined by dividing the energy content of hydrogen, based on its HHV, by the electrolyzer's efficiency. This relationship is expressed in Equation (19):

$$C_{\mathrm{ONS}_E} = \frac{100 \cdot Q \cdot \mathrm{HHV}_{H_2}}{\eta_E\,\%}, \tag{24}$$

where Q represents the hydrogen mass flow rate in kilograms per hour (kg/h), $\mathrm{HHV}_{\mathrm{H2}}$ is the higher heating value of hydrogen, which is 39.4 kWh/kg [37], and η_E % denotes the efficiency of the electrolyzer. This approach ensures a comprehensive and accurate assessment of the system's techno-economic performance [18, 29].

2.4.3 | LCOH

The LCOH is computed in HOMER by dividing the difference between the total annualized cost ($C_{\mathrm{ann,tot}}$) and the annual revenue from electricity sales ($C_{\mathrm{electricity}}$) by the total annual hydrogen production (M_{hydrogen}), as expressed in Equation (20) [29].

$$\mathrm{LCOH} = \frac{C_{\mathrm{ann,tot}} - C_{\mathrm{electricity}}}{M_{\mathrm{hydrogen}}}. \tag{25}$$

In this study, the green hydrogen-producing plant operates off-grid, rendering $C_{\mathrm{electricity}}$ zero since no electricity is sold to the grid. Notably, HOMER lacks economic parameters for compressors and dispensers, as these are not modeled as components or loads. However, more accurate LCOE and LCOH estimates can be calculated separately and incorporated into $C_{\mathrm{ann,tot}}$ [29]. Detailed technical and economic parameters related to all components of the considered green hydrogen production plant are mentioned in Table 2.

3 | Results and Discussions

The hydrogen production plant in this study was designed to meet an initial daily demand of 100 kg (with scalability potential) and modeled using HOMER to simulate all possible configurations of a wind-PV hybrid system with battery storage, specifically tailored to Umgababa's parameters. The techno-economic analysis of these configurations—PV-battery (Case 1), wind-battery (Case 2), and hybrid PV-wind-battery (Case 3)—yielded critical insights into system performance and feasibility, with optimization results (detailed in Table 3) evaluated against key metrics including NPC, LCOH, and Energy (LCOE), renewable fraction (RF), and excess electricity production.

3.1 | Comparative Performance of the Three Configurations

The NPC values for the three systems (Figure 2) were remarkably close, with the wind-battery system (Case 2) exhibiting the lowest NPC at $5,279,048, followed by the hybrid PV-wind system (Case3) at $5,281,333, and the PV-battery system (Case 1) at $5,286,686. The marginal differences suggest that all three systems are economically competitive, with the wind-battery system holding a slight edge due to lower capital and operational costs associated with wind energy infrastructure in the studied location. This aligns with findings by [18], who noted the cost-effectiveness of wind energy in coastal regions of South Africa due to consistent wind resources.

The hybrid PV-wind system (Case 3) demonstrated the lowest LCOH at 8.94/kg and highest annual hydrogen production (45.8 tons/year), outperforming standalone systems. outperforming both the PV—battery and wind—battery systems, which recorded $11.2/kg. This reduction in LCOH for the hybrid system can be attributed to the synergistic effect of combining solar and wind resources, which mitigates intermittency and maximizes hydrogen production efficiency. The hybrid system's ability to leverage complementary generation profiles (solar peaks during daylight and wind peaks at night) ensures a more stable and continuous energy supply to the electrolyzer, thereby optimizing hydrogen output. This finding is consistent with studies by Al-Buraiki and Al-Sharafi [22], who highlighted the role of hybrid systems in minimizing LCOH through improved resource utilization.

TABLE 3 | Optimized green hydrogen powered systems for three cases (PV, wind, and wind-PV with battery parameters related to all components.

Case	1 - PV	2 - Wind	3 - PV-Wind
PV-capacity (kW)	2000	0	500
Wind turbine quantity	0	1	10
Electrolyzer capacity (kW)	1000	500	500
Battery quantity	24	24	24
Hydrogen tank size (kg)	300	300	300
Converter capacity (kW)	10.00	322.19	218.13
Annual hydrogen production (ton/yr)	36,605	36,650	45,767
NPC ($)	5,286.69	5,279.05	5,281.33
Ren frac (%)	100	100	100
Excess elec (%)	46.71	79.30	97.46
LCOE $/kg	99.69	99.55	99.59
LCOH $/kg	11.20	11.20	8.94

Abbreviations: LCOE, levelized cost of electricity; LCOH, levelized cost of hydrogen; NPC, net present cost; PV, photovoltaic; Ren Frac, renewable fraction.

This finding is consistent with Safak and Çiçek [14], who reported that RES integration significantly reduces external dependency and enables surplus energy trading, thereby enhancing financial gains. It is also consistent with Ahmad et al. [4], who highlighted the synergistic effects of hybrid systems in minimizing costs through improved resource utilization. The high excess electricity (97.5%) underscores the scalability potential, a key advantage noted in techno-economic analyses by Naqvi et al. [5].

The LCOE values were nearly identical across all systems (~$99.6/kg), indicating that the cost of electricity generation was not a differentiating factor in this analysis. However, it is noteworthy that the hybrid system achieved this similarly while producing significantly higher hydrogen volumes (45.8 tons/year vs. ~36.6 tons/year for standalone systems), highlighting its superior efficiency.

All systems achieved a 100% RF, confirming their sustainability. However, the hybrid system exhibited the highest excess electricity (97.5%), followed by the wind-battery (79.3%) and PV-battery (46.7%) systems. While excess electricity reflects underutilised potential, it also indicates opportunities for scaling hydrogen production or integrating additional storage solutions. The hybrid system's high EE suggests that further optimization (e.g., larger electrolyzer capacity or grid export) could enhance economic returns.

3.2 | Monthly Energy and Hydrogen Production Analysis

For a PV-Battery System (Case 1), solar energy production peaked in the summer months (December–January), aligning with Durban's high solar irradiance (Figure 7a). Hydrogen production mirrored this trend, but the system struggled during low-irradiance periods (June–July), necessitating battery backup. Wind energy production was more consistent year-round for the Wind-Battery System (Case 2), with peaks in October (Figure 7b). This consistency translated into stable hydrogen output, although the system's lower electrolyzer capacity (500 kW vs. 1000 kW in Case 1) limited total production.

3.2.1 | Monthly Energy Production Analysis: Electrical Power Generation

The monthly electricity generated by the PV-battery system (Figure 7a), the Wind-battery system (Figure 7b) and the hybrid PV-wind battery system (Figure 7c) under the climatic conditions of Durban. This was carried out to ascertain the overall electricity production and the balance of energy supply by individual source to meet the demand for green hydrogen production.

Figure 7a shows clear seasonal trends, peaking during the summer months (September–January) due to the longer daylight hours and higher solar irradiance. Notwithstanding the subtropical climate zone of Durban, the solar potential remains moderate even in winter, contributing consistently to the electricity generation by the system. Figure 7b depicts the electricity generation by the wind-battery system trends to complement the PV-battery system, often peaking during cooler months (August -December), where wind speeds are generally high. Figure 7c shows the electricity generated by the hybrid PV-wind battery system, which complements smoothing out seasonal and daily fluctuations in renewable energy availability. It also maintained a consistent energy supply, which was crucial for uninterrupted green hydrogen production and reliable power support for the electrolyzer.

3.2.2 | Monthly Energy Production Analysis: Hydrogen Generation

In this section, the energy produced by the PV-battery system (Figure 8a), the Wind-battery system (Figure 8b) and the hybrid PV-wind battery system (Figure 8c) were evaluated for the green hydrogen production. It was deduced that the hybrid PV-wind battery system offered technically and economically viable, consistent renewable electricity in the coastal region of South Africa. This makes it highly sustainable for powering green hydrogen production and supporting the just energy transitions.

The monthly hydrogen generation data presented in Figure 8a–c indicate the performance of three renewable power systems: PV-battery, wind-battery, and hybrid PV-wind-battery configurations. The solar irradiance-driven hydrogen production from the PV-battery system (Figure 8a) fluctuates according to seasonal sunshine patterns, for example, higher generation in sunnier months (e.g., summer) and lower output during winter due to insufficient daylight conditions and weather fluctuations. In contrast, the wind-battery system (Figure 8b) has fewer predictable monthly fluctuations, reflecting the site-specific and seasonal nature of wind resources, whereby generation is most prevalent in months of strong winds (e.g., spring or autumn) but is still affected by short-term weather fluctuations. The hybrid PV-wind-battery system (Figure 8c) demonstrates enhanced performance, synergizing the strengths of both technologies to offset

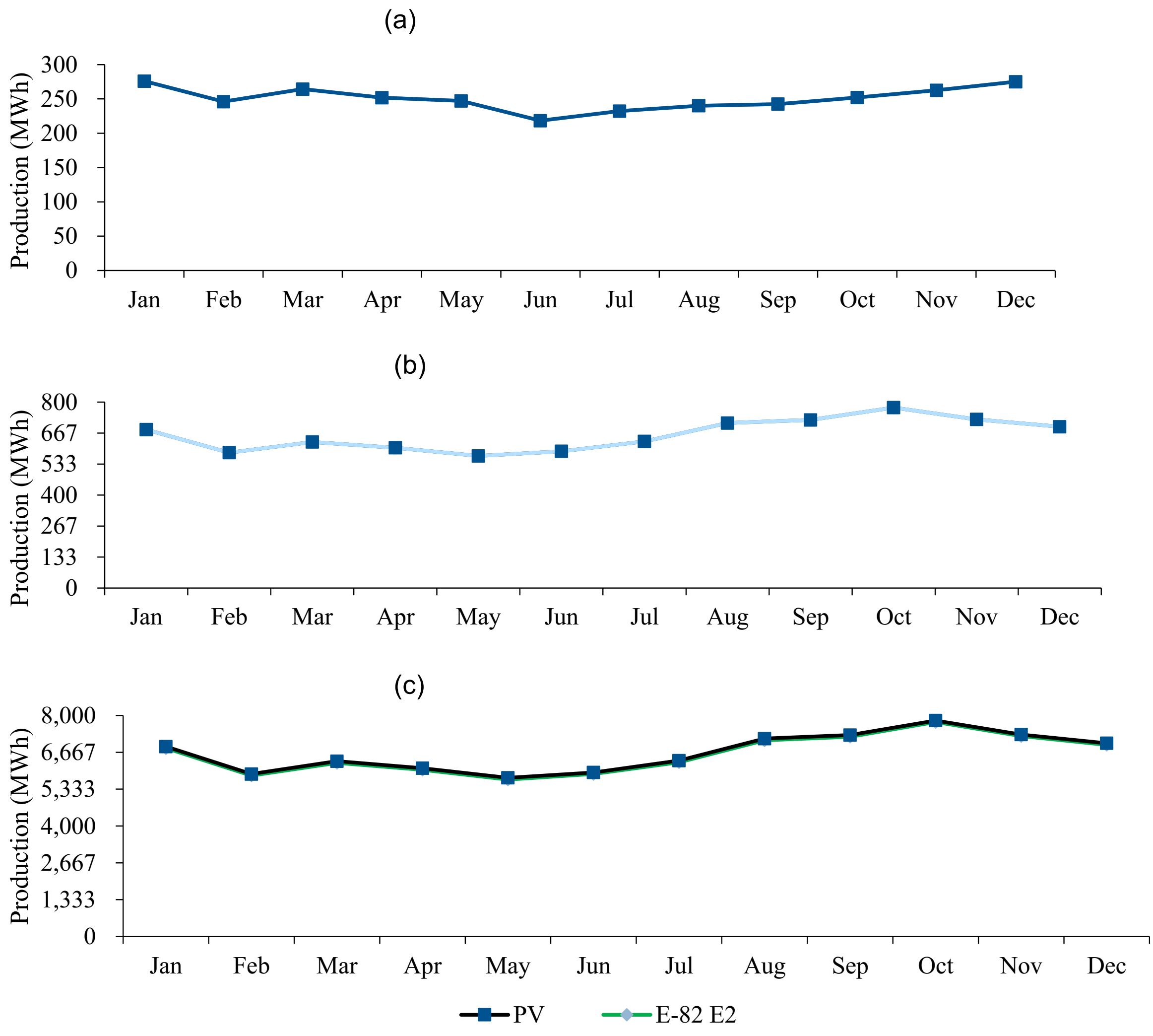

FIGURE 7 | Monthly Electrical power generation by (a) PV-battery system, (b) Wind-battery system, and (c) hybrid PV-Wind system.

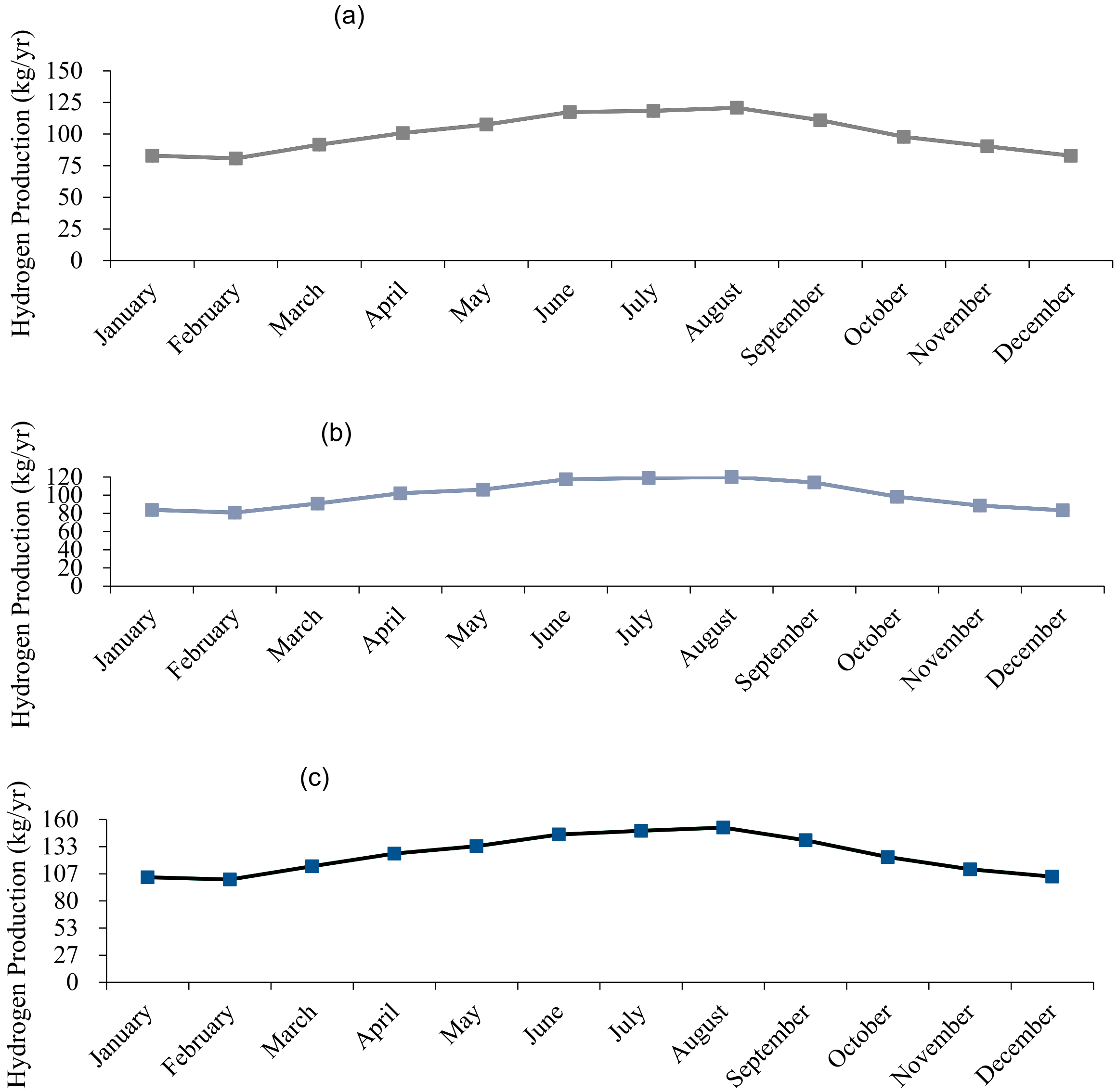

FIGURE 8 | Monthly hydrogen generation by (a) PV-battery system, (b) wind-battery system, and (c) hybrid PV-wind-battery system.

intermittent solar deficits balanced by wind surpluses, and vice versa, resulting in more stable and efficient hydrogen production throughout the year. This demonstrates the efficiency and reliability of the hybrid system compared to single configurations for sustainable hydrogen production. Hybrid PV-Wind System (Case 3) successfully incorporated the strengths of both resources, solar energy being the dominant one during daytime and assistance provided by wind energy during the evening and over cloudy days. This synergy resulted in the highest annual hydrogen production (45.8 tons/year), demonstrating the system's robustness against variability (Figure 8c).

3.2.3 | Excess Electricity Generated

The excess electricity, being referred to as the amount of power generated that is not consumed or utilized by the load or stored by the battery due to low demand, has the potential for grid export. Analyzing the efficiency and the excess energy by the PV-battery system (Figure 9a), the Wind-battery system (Figure 9b) and the hybrid PV-wind battery system (Figure 9c) provided insight for sizing the electrolyzer, the green hydrogen production and storage.

The graphs (Figure 9a–c) illustrate three battery-integrated renewable energy systems, PV, Wind, and Hybrid Wind-Battery; each displaying distinct electricity generation patterns. The PV system (Figure 9a) shows highly variable output with short-duration peaks, on average, 30%–65% excess electricity generated monthly. Whereby this electricity could be utilized or redirected to the grid. The Wind-Battery system (Figure 9b) also demonstrated more consistent generation, where an average of about 25%–50% excess electricity was generated monthly. This also strengthened the turbine sizing and battery storage adequacy. On the other hand, the Hybrid Wind-Battery system (Figure 9c) exhibits the most stable and high-capacity production. It was deduced that the hybrid system can potentially produce, on average, 75%–90% excess electricity. This allows the scalability of green hydrogen production, increasing the capacity, battery enhancement, load capacity and feeding to the national grid. This becomes very critical for the selection of battery storage in mitigating the intermittency of RESs, with

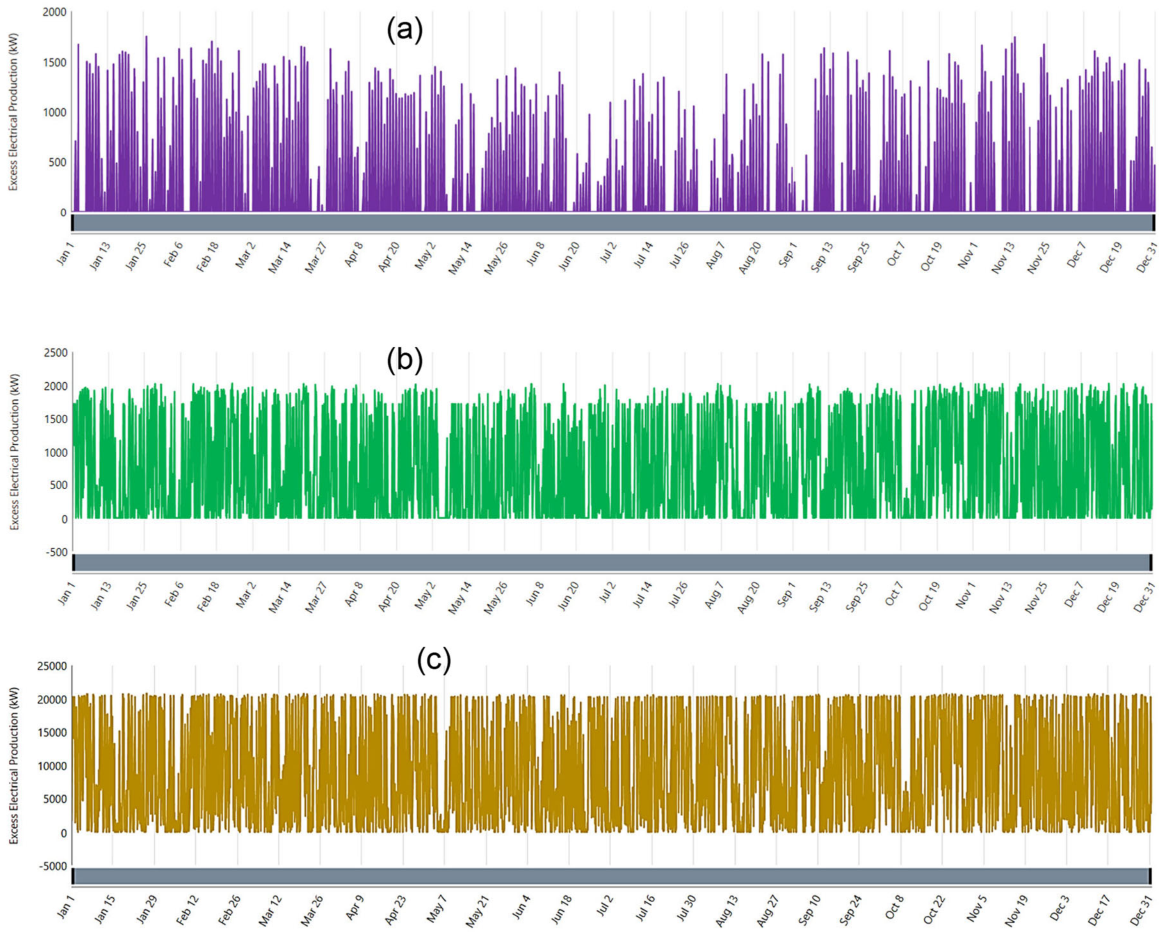

FIGURE 9 | Monthly Excess Electrical Production by (a) PV-battery system, (b) wind-battery system, and (c) hybrid PV-wind-battery system.

hybrid systems offering the most robust and reliable approach to clean energy generation.

The hybrid PV-wind system's 97.5% excess electricity generation presents significant opportunities for scalability, grid integration, and sector coupling, as demonstrated in recent studies. Like Erding et al. [17], who used excess energy for hydrogen production and FCEV charging, this system could leverage surplus electricity for additional hydrogen production, grid export, or desalination, enhancing economic viability. The high excess energy could also power reverse osmosis (RO) desalination units, as shown in Romo et al. [15], where RO systems accomplished 47% recovery ratios with a LCOH competitive with freshwater electrolysis, requiring 2.2–2.5 kWh/m^3. For instance, producing 100 kg of hydrogen daily would require about 50 kWh more each day, which can be handled by the system's excess capacity. This dynamic balancing of supply and demand further supports resilience-oriented frameworks for green buildings, as highlighted in recent research by Çiçek [10].

3.2.4 | Hydrogen Storage Analysis

The hydrogen tank storage level reflects the balance between hydrogen production via electrolysis and the consumption or export to the grid. The three renewable energy configurations (Figure 10a), hybrid PV-wind-battery system, (Figure 10b) PV-battery system, (Figure 10c) wind-battery system, exhibited distinct patterns in hydrogen storage dynamics due to differences in energy generation profiles (Figures 7 and 8) and excess electricity availability (Figure 9). A comparison of tank levels in Figure 10 illustrates notable differences between hybrid PV-wind, PV-battery, and wind-battery systems. The hybrid PV-wind system (Figure 10a) illustrates maximum stability and consistency among tank levels on different days of the year, as represented by evenly oscillating color change, indicating dynamic but controlled storage levels. In contrast, the PV-battery system illustrates erratic and fluctuating indicating more intermittent storage capabilities. The wind-battery system exhibits periodic fluctuations, highlighting its intermittent storage capacity. In contrast, the hybrid PV-wind system emerges as the optimal solution, delivering enhanced stability and more reliable energy storage performance.

3.3 | Economic and Environmental Trade-Offs

The economic performance of the renewable energy systems powering the seawater electrolyzer for green hydrogen production was assessed through key metrics such as NPC and LCOH. As shown in Table 3, in comparing the three configurations, the PV-Wind system (Case 3) with the lowest LCOH of 8.94 $/kg was considered the most cost-effective, whereby a detailed economic performance (NPC) was carried out

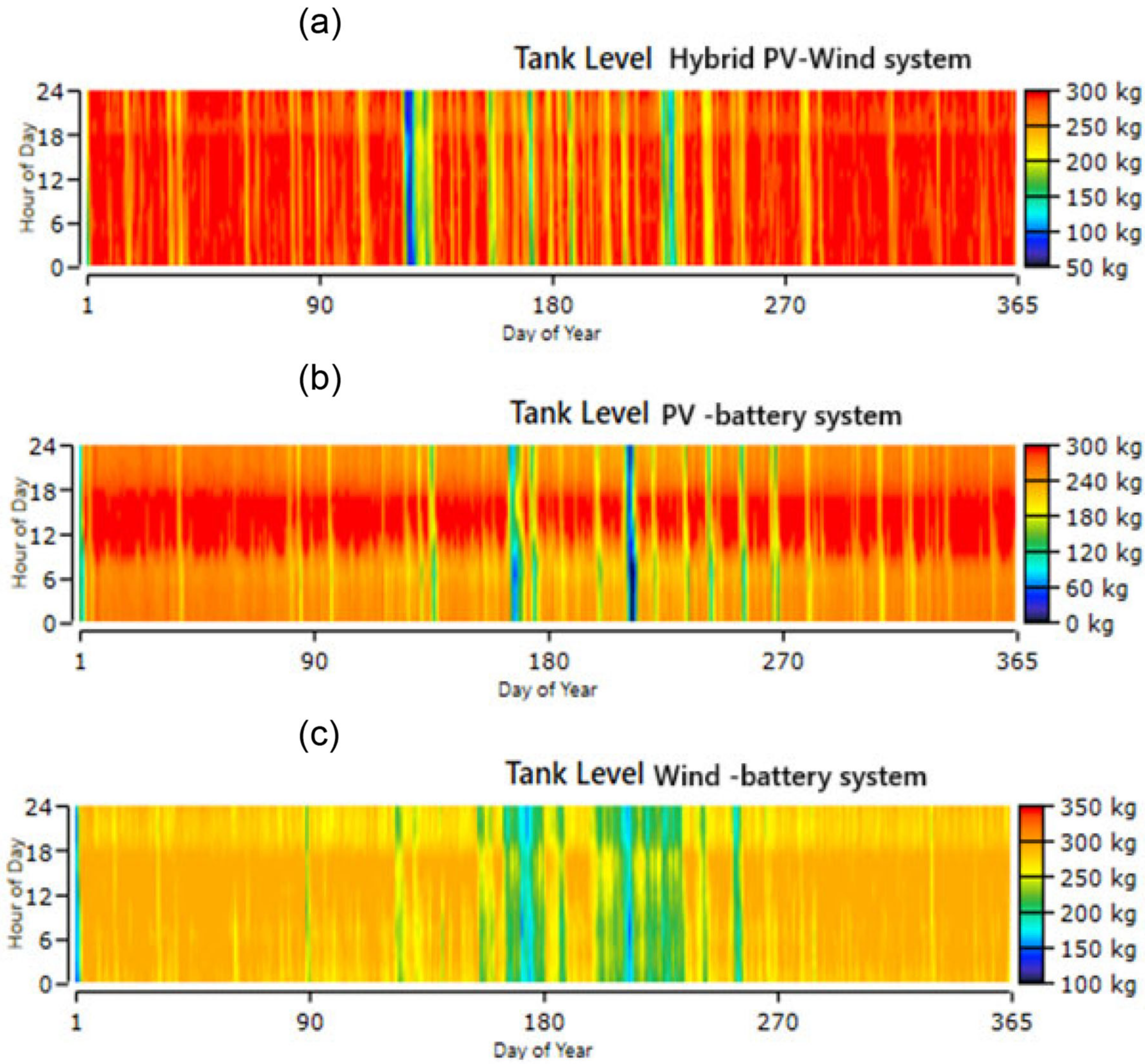

FIGURE 10 | Hydrogen tank storage level for different renewable configurations for (a) hybrid PV-wind-battery system, (b) PV-battery system, and (c) wind-battery system.

(Figure 11). This made it an economically feasible configuration system with scalability for green hydrogen production.

The cost breakdown in Figures 11 and 12 reveals that the hybrid PV-wind system's capital costs are dominated by PV arrays and electrolyzers, while replacement and O&M costs follow similar trends. Salvage value from components like converters partially offsets expenses. The annualized cost summary shows net expenditures are balanced by salvage returns, confirming the system's economic viability despite high initial investments.

3.3.1 | Environmental Impact

All three alternative power configurations (PV-battery, wind-battery, and hybrid PV-wind-battery) achieved a 100% RF, zero operational emissions, and ensured their sustainability in harmony with worldwide efforts towards the reduction of carbon footprints and the battle against climate change. The hybrid system's 100% RF and zero operational emissions are complemented by its potential to integrate desalination, as shown in Romo et al. [15], where multieffect distillation (MED) and reverse osmosis (RO) systems achieved theoretical efficiencies of 78%–95%."

While all the systems reduced emissions, the hybrid PV-wind system had superior environmental benefits, generating 45.8 tons/year of hydrogen (Table 3), significantly reducing the reliance on fossil fuels in uses like transport and industry, in line with South Africa's Hydrogen Society Roadmap vision goals [38]. The hybrid system significantly reduced intermittency by harnessing the complementary nature of solar and wind resources, ensuring a stable energy supply while decreasing dependence on backup fossil fuel generation. Specifically, the hybrid configuration demonstrated the highest excess electricity output (97.5%), with untapped potential for increasing hydrogen output or integrating additional storage components, as represented in Figures 8 and 9.

Additionally, Figure 10 reflected the hybrid system's constant hydrogen storage levels, as compared to changing levels of the PV-battery and cyclic variations of the wind-battery systems. This complementarity of daytime solar and nighttime wind power provides constant production while improving system

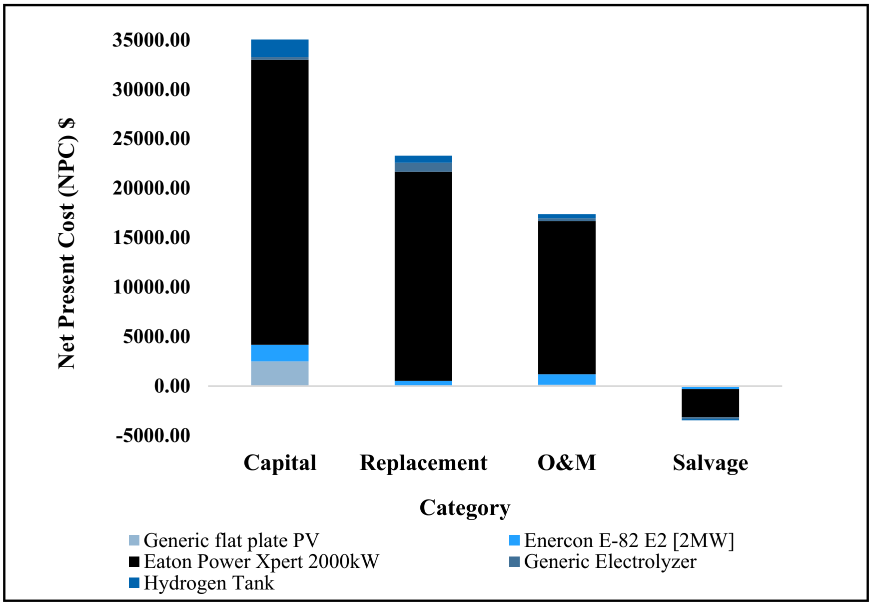

FIGURE 11 | Net present cost (NPC) summary of various components for the PV-Wind hybrid system.

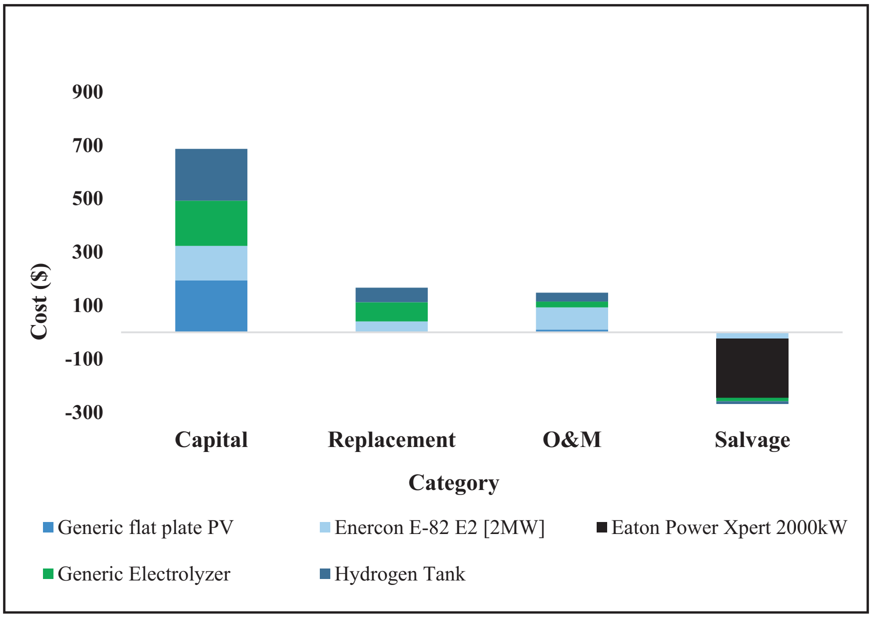

FIGURE 12 | Annualized cost summary of various components for the PV-wind hybrid system.

robustness with less battery reliance. The scalability of the hybrid system, backed by its high excess electricity, implies potential for increased hydrogen production or grid exports, further increasing its environmental footprint. Future policy support could expedite large-scale deployment, promoting South Africa's low-carbon economy transition. The findings of this study support scientific understanding of green hydrogen production, advancing a fair transition and aligning with crucial Sustainable Development Goals (SDGs), such as clean and affordable energy (#7), quality education (#4), clean and accessible energy (#7), innovation in sustainable infrastructure (#9), and climate action (#13). Thus, the hybrid PV-wind-battery system emerges as an economically and environmentally favorable solution, a cornerstone for South Africa's green hydrogen future.

3.3.2 | Optimal System Selection

The wind-battery system (Case 2) is the most economical in terms of NPC, while the hybrid PV-wind system (Case 3) emerges as the most optimal configuration overall due to: Lower LCOH ($8.94/kg vs. $11.2/kg), which is critical for commercial viability. The hydrogen production (45.8 tons/year vs. ~36.6 tons/year), enabling scalability and resource complementarity, reducing reliance on batteries and enhancing system reliability. This conclusion is supported by Gökçek and Kale [29], who emphasized the techno-economic advantages of hybrid systems in regions with balanced solar/wind resources, such as Durban.

4 | Conclusions

This study conducted a techno-economic analysis of solar-wind hybrid systems for green hydrogen production in South Africa's KZN province using HOMER software. Three configurations: PV-battery (Case 1), wind-battery (Case 2), and hybrid PV-wind-battery (Case 3) were evaluated. The hybrid PV-wind-battery system emerged as the most optimal configuration, outperforming standalone systems by achieving the lowest LCOH at $8.94/kg (compared to $11.2/kg for PV or wind alone) and the highest annual hydrogen output (45.8 tons/year). Its ability to leverage complementary solar (daytime) and wind (nighttime) resources to ensure a stable energy supply. The hybrid system's NPC of $5,281,333 was marginally higher, making it more resilient and scalable. All the systems achieved 100% renewable energy, with the hybrid system exhibiting the highest excess electricity (97.5%). This offers future opportunities for the scalability of green hydrogen and energy storage, as well as integrating with a grid system.

Future research directions should explore dynamic optimization of electrolyzer sizing and smart control strategies for excess energy management, as suggested by Naqvi et al. [5], alongside real-time system performance monitoring.

While this study is simulation-based, future work should also include experimental validation or calibration with field data leveraging methodologies like Romo et al. [15]'s operational data recovery framework or resilience-oriented energy management approaches (Çiçek [10]), to confirm the reliability of findings for real-world applications. Additionally, incorporating desalination energy demand and cost modeling is critical to address seawater electrolysis challenges in water-scarce regions, building on insights from multi-energy systems (Safak and Çiçek [14]). Advanced energy management strategies, such as bidirectional grid interaction, could further improve system efficiency and scalability (Safak and Çiçek [14]). Integrating long-term storage technologies and policy support could further enhance scalability, mirroring global advancements in green hydrogen production as reported by Ahmad et al. [4].

Author Contributions

Sandile Mtolo: conceptualization, methodology, software (HOMER), validation, formal analysis, investigation, resources, data curation, writing – original draft preparation, visualization. **Sudesh Rathilal:** conceptualization, validation, writing – review and editing, supervision, funding acquisition. **Nomcebo Happiness Mthombeni:** validation, supervision. **Emmanuel Kweinor:** conceptualization, software (HOMER), validation, resources, writing – review and editing, visualization, supervision, funding acquisition, methodology. **Katleho Moloi:** software (HOMER), validation, supervision. All authors have read and agreed to the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data presented in manuscript and also available upon request.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

The authors declare the use of AI tool such as HOMER software was used for the data analysis and the use of Grammarly software was employed to improve the grammar of the write-up. After using these tools/service, the authors reviewed and edited the content as needed and take full responsibility.

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